KEREM FARUK ÖZTÜRK

Electrical & Electronics Engineering | Digital Hardware Design & Verification | AI Hardware & Security | Python, C#/.NET
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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering undergraduate at Istanbul Bilgi University focusing on AI hardware: digital design and verification of neural network accelerators. Experienced in backend software engineering through payment-system integrations at VakıfBank and AI code evaluation at Outlier (Scale AI). Since November 2026, building a small neural network accelerator in Verilog (FPGA, then Tiny Tapeout) with an open-source lint, simulation, and formal verification flow.

EDUCATION

Istanbul Bilgi University 2023 – Expected Jul 2028
B.Sc. in Electrical and Electronics Engineering (100% English Instruction) Istanbul, Türkiye

PROFESSIONAL EXPERIENCE

VakıfBank Jul 2026 – Sep 2026
Software Development Intern – Payment Systems Application Development (Head Office) Istanbul, Türkiye

- **Virtual POS Merchant Integration Application (.NET / C#):** Developed a comprehensive, functional .NET solution enabling external merchants to execute live sale, transaction inquiry, cancellation/void, and refund operations, including scenario-based sample implementations and starter integration templates.
- **Shopify Payments App & Payments Extension (ASP.NET Core / PostgreSQL):** Contributed to a Payments App and Extension bridging Shopify merchants with VakıfBank's virtual payment infrastructure, working on .NET backend services, REST API integrations, EF Core, and PostgreSQL data access across payment, 3D Secure, inquiry, cancellation, and refund flows.

Outlier (Scale AI) Nov 2024 – May 2026
Python Code Reviewer / AI Code Evaluator Remote

- Evaluated AI-generated Python solutions for functional correctness, algorithmic efficiency, and code quality.
- Conducted manual code inspections to detect logical errors, edge-case failures, performance bottlenecks, unsafe coding practices, and potential security vulnerabilities, authoring structured technical feedback reports.

INDEPENDENT TECHNICAL PROJECTS

Neural Network Inference Accelerator (FPGA & Tiny Tapeout) In Progress (since Nov 2026)
Verilog | cocotb | Icarus Verilog | Verilator | Yosys | SymbiYosys | Python | FPGA | Tiny Tapeout (GF180MCU)

- Designing a compact INT8 neural network inference accelerator in Verilog for handwritten-digit classification: multiply-accumulate (MAC) datapath with externally streamed weights, control FSM, ReLU, and argmax stage.
- Verification: bit-exact comparison against a Python fixed-point model (cocotb), plus lint and SymbiYosys formal checks.
- Planned milestones: FPGA bring-up with ready/valid interfaces; measured comparison of variants (weight bit-width, serial vs. parallel MAC) on area, timing, and accuracy; Tiny Tapeout submission in 2027, followed by silicon measurement.

RTL-SecCheck: Auditing AI-Generated Hardware Verification In Progress (since Nov 2026)
Python | Yosys | SymbiYosys | cocotb | LLM APIs

- Phase 1 (ongoing): lint, cocotb, and SymbiYosys check flow for the accelerator RTL, logging AI output and defects caught.
- Phase 2 (2027): measuring whether AI-generated verification properties catch known bugs in open designs with documented defects (Hack@DAC, OpenTitan), using mutation testing to flag vacuous proofs.

48 V to 12 V Intermediate Bus Converter (Bench-Scale Model) 2027
LTspice | KiCad | Synchronous Buck | Digital Control | Bench Measurement

- Bench-scale model of the data-center bus stage that powers AI hardware: digitally controlled synchronous buck (first prototype 24 V to 5 V), with efficiency, load-step, and loop-gain measurements compared against LTspice.

TRAINING & PROGRAMS

Türkiye Siber Vatan Program – AI-Supported Cybersecurity Talent Program (2027 Cohort) Nov 2026 – Present
Republic of Türkiye Ministry of Industry and Technology & Regional Development Agencies Türkiye

- Admitted to the national cybersecurity and AI security talent program; online training phase in progress, followed by penetration testing, AI-for-cybersecurity, and AI CTF stages.

Digital University Platform – Artificial Intelligence and Modern Software Development Nov 2026 – 2027
Council of Higher Education of Türkiye (YÖK) Online

- AI School module on AI-assisted code development, software architecture, requirements, and testing.

Google Türkiye – Artificial Intelligence and Technology Academy One-Year Training Program
Selected Scholarship Recipient Türkiye

- Completed training on LLM fundamentals and AI-assisted development; built an AI-integrated web app in a bootcamp team.

TECHNICAL SKILLS

- **Hardware Design & Verification (Developing):** Verilog, cocotb, Icarus Verilog, Verilator, Yosys, SymbiYosys, FPGA Design
- **Software Development & Backend:** Python, C#, ASP.NET Core, SQL, REST APIs, OOP, Virtual POS & 3D Secure Integrations
- **Tools & Practices:** PostgreSQL, EF Core, Git/GitHub, Docker, Linux, Postman, DBeaver, Code Review, Security Inspection
- **Languages:** Turkish (Native), English (Professional Working Proficiency)